

Post-irradiation microstructural examination of EUROFER-ODS steel irradiated at 300°C and 400°C

M. Klimenkov*, U. Jäntschi, M. Rieth, M. Dürschnabel, A. Möslang, H.C. Schneider

Karlsruhe Institute of Technology, Institute for Applied Materials, Hermann-von-Helmholtz-Platz 1, 76344 Eggenstein-Leopoldshafen, Germany

ARTICLE INFO

Article history:

Received 28 May 2021

Revised 23 July 2021

Accepted 17 August 2021

Available online 4 September 2021

Keywords:

EUROFER-ODS

Neutron irradiation

Dislocation loops

Burgers vector

TEM

ODS

ABSTRACT

The effect of neutron irradiation on the microstructure of EUROFER-ODS alloy reinforced with an addition of 0.5% Y_2O_3 was analyzed by transmission electron microscopy. The irradiation of up to 16.2 dpa damage dose at 300°C and 400°C was performed in Petten High Flux Reactor. This study focuses on the imaging, identification, and quantitative analysis of interstitial loops and voids located on ODS particles. The nano-sized loops and point defects with $b\frac{1}{2}\langle 111 \rangle$ Burgers vectors are preferentially located near structural defects such as the line dislocations or impurities in the material irradiated at 300°C. Homogeneously distributed loops having Burgers vectors of $b\frac{1}{2}\langle 111 \rangle$ and $b\langle 100 \rangle$ types were found in the material irradiated at 400°C. The obtained quantitative data on radiation-induced defects such as size, number density and relaxation volume were compared with EUROFER97 irradiated at the same conditions. It was found that the occurrence of radiation induced defects depends on the local number density and the size distribution of the ODS particles and not only on the irradiation damage and temperature.

© 2021 Elsevier B.V. All rights reserved.

1. Introduction

Ferritic-martensitic oxide-dispersion-strengthened (ODS) steels are promising candidates for numerous applications in nuclear and fusion technology due to their superior radiation resistance properties [1,2]. The importance of this topic is shown by the fact that dozens of ODS alloys have been developed, optimized and tested over the last two decades, and yet this has not lost any of its relevance [3–5]. The most recent development have focused on the fabrication and characterization of austenitic ODS alloys [6]. An ODS variant of EUROFER97 was developed with the purpose of further increasing the operating temperature and thus the efficiency of advanced blanket concepts for fusion reactors.

Neutron irradiation and ion implantation are both experimental techniques applied for the verification of radiation resistance of ODS steels [4,7–9]. The advantages of ion implantation are relatively low costs, short irradiation time, providing non-radioactive samples [4,9,10] as well as the ability to accumulate significant damage of several tens of dpa in the acceptable time scale [5]. However, the influence of ion and neutron irradiation on the microstructure can be quite different, since the damage rate appears as an important factor for defect formation [11]. For this reason, the study of microstructure of neutron irradiated ODS alloys is of great importance to understand their structural evolution

under application conditions. It should be mentioned that enormous technical efforts and costs for neutron irradiation experiments make the published work on neutron irradiation of ODS materials quite rare [8,12].

Since neutron irradiated EUROFER-ODS steels exhibit in a wide temperature window superior tensile and fatigue properties compared to casted steels like for example EUROFER97 [13], knowledge of the defect formation mechanisms is important to understand the effect of ODS particles on radiation resistance and the relationship between irradiation modified microstructures and mechanical properties. Highly dispersed oxide particles serve as trapping sites for vacancies and self-interstitial atoms, preventing the formation and coarsening of radiation-induced defects that lead to degradation of mechanical properties. Most studies of irradiated ODS materials are focused on demonstrating the stability of ODS particles [7,14,15] and to a lesser extent, on the formation of radiation-induced defect [8,16]. Our work intends to fill the current knowledge gap and contribute to a better understanding of the radiation resistance mechanism of ODS alloys.

The experimental results of this study reveal the microstructural properties of radiation-induced defects formed at 300°C and 400°C. These should complete the characterization of the of the first generation EUROFER97 and EUROFER-ODS alloys irradiated within the SPICE program, which are already published earlier [3,12]. In addition to a detailed defect analysis, it is also shown that the formation of radiation-induced defects depends not only on the irradiation temperature but also on the local distribution

* Corresponding author.

E-mail address: michael.klimenkov@kit.edu (M. Klimenkov).

Table 1

|b•g| values for [111] beam direction and for g=[020] near [102] zone axis.

$g \setminus b$	$\frac{1}{2}[111]$	$\frac{1}{2}[\bar{1}\bar{1}\bar{1}]$	$\frac{1}{2}[\bar{1}\bar{1}1]$	$\frac{1}{2}[11\bar{1}]$	[001]	[010]	[100]
$[\bar{1}\bar{1}0]$	0	1	0	0	0	1	1
$[\bar{1}01]$	0	1	1	1	1	0	1
$[01\bar{1}]$	0	0	1	1	1	1	0
[020]	1	1	1	1	0	2	0

of ODS particles (i.e., size and local number density). The temperature influence on the formation of radiation-induced defects will be discussed with a regard to these previously published results.

2. Experimental

The EUROFER-ODS alloy examined in this study was fabricated by mechanical alloying of EUROFER97 powder with addition of 0.5 wt% Y_2O_3 in industrial ball mills the type attritor (Plansee, Austria). The EUROFER97 powder of 8.9 wt% Cr, 1.1 wt% W, 0.2 wt% V, 0.14 wt% Ta, 0.42 wt% Mn, 0.06 wt% Si, 0.11 wt% C composition was produced by inert gas atomisation (HC-Starck, Laufenburg, Germany). Hot Isostatic Pressing (HIP) was used as a consolidation process for the production of bars with a diameter of 60 mm and a length of 300 mm. This material was available for numerous laboratories in Europe and has been comprehensively characterized [17]. The tensile and Charpy specimens for irradiation program were prepared from the hiped material without further thermal treatment. The neutron irradiation was carried out in the Petten High Flux Reactor (HFR) within the framework of the HFR irradiation project Phase IIb (SPICE) with a nominal dose of 16.3 ± 1.9 dpa at the targeted irradiation temperatures of 250°C, 300°C, 350°C, 400°C and 450°C. [18]. The content of transmutation-induced helium was estimated to <10 appm. The present work reports the microstructure of EUROFER-ODS irradiated at 300°C and 400°C targeted temperature. The average temperature measured during irradiation was $(293 \pm 8)^\circ\text{C}$ and $(392 \pm 8)^\circ\text{C}$, accordingly. Since the target and average temperatures are within the error range, we use the target temperature when discussing the results. The microstructure of the material irradiated at 250°C, 350°C and 450°C targeted temperatures was investigated and published by Klimenkov et al. [12].

The TEM specimens were prepared by electrochemical etching. Disks with a thickness of 0.2 to 0.3 mm and 3 mm in diameter were thinned by electropolishing in a TENUPO V device using $H_2SO_4 + 80\% CH_3OH$ electrolyte. In the next step, disks of 1 mm diameter were punched with the hole in the middle to minimize the quantity of the radioactive sample and thus the irradiation exposure and optical distortion caused by the magnetic field in TEM. TEM investigations were performed using a FEI Tecnai 20 FEG analytical microscope with an accelerating voltage of 200 kV. Preparation and analysis of the irradiated samples was done in the controlled area of the Fusion Materials Lab at KIT.

The Burgers vector of dislocation loops was determined by weak beam dark field (WBDF) imaging with low indexed $g(\text{ng})$ vectors near the [100] and [111] zone axes. The tilt experiments were performed using a standard FEI double tilt holder. The corresponding visibility criteria of the dislocation loops for different g-vectors are listed in tables 1 and 2.

3. Results

3.1. Microstructure of unirradiated material

The EUROFER-ODS alloys with 0.3 % and 0.5 % Y_2O_3 has been subjected to an extensive examination in the past [12,19–21]. Their microstructure consists of ferritic grains with an average size of several micrometers [12]. The presence of two types of precipitates

Table 2

|b•g| values for all g vectors in orientations close to the [100] zone axis. The loops of defined Burgers vector are drawn with corresponding colours in the Figs. 7 and 8.

$g \setminus b$	$\frac{1}{2}[111]$	$\frac{1}{2}[\bar{1}\bar{1}\bar{1}]$	$\frac{1}{2}[\bar{1}\bar{1}1]$	$\frac{1}{2}[11\bar{1}]$	[001]	[010]	[100]
[011]	1	1	0	0	1	1	0
[01 $\bar{1}$]	0	0	1	1	1	1	0
[002]	1	1	1	1	2	0	0
[020]	1	1	1	1	0	2	0
color	blue	blue	red	red	yellow	green	—

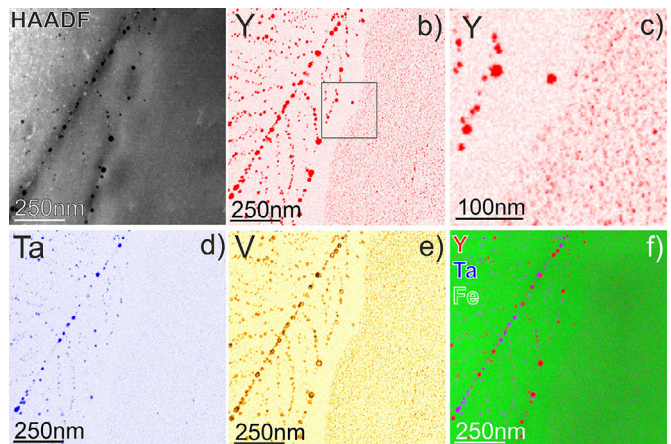


Fig. 1. Analytical examination of unirradiated EUROFER-ODS. The HAADF image of analyzed area is shown in part (a) and the maps of the corresponding elements imaged with reverse contrast are shown in parts (b–e). Part (c) shows an enlarged section from part (b) which is marked with a square. The part (f) represents composition of Y (red), Ta (blue) and Fe (green) maps.

has been reported: Micrometer-sized $(Cr,Fe)_{23}C_6$ carbides, located mostly at the grain boundaries, and finely dispersed nanometer-sized ODS particles, composed basically of Y_2O_3 [12,19]. In general, the grain structure and distribution of ODS particles of the alloy with 0.5% Y_2O_3 are very similar to that of the EUROFER-ODS alloy containing 0.35% Y_2O_3 [20,21]. Despite this comprehensive knowledge, we present new aspects of the composition of ODS particles that have not been published previously.

Analytical investigations demonstrating spatial distribution of ODS particles in both areas are shown in Fig. 1. The HAADF image of investigated area is given in Fig. 1a, while the elemental maps using Y-L and Ta-M, V-K and Fe-K EDX lines of the marked area are presented in the parts (b)–(e) respectively. The size distribution of the ODS particles shows significant variations for different areas. In the "coarse" area, the size of ODS particles varied from 5 nm to 25 nm with number density that varied for different locations from $2.5 \cdot 10^{20} \text{ m}^{-3}$ to $5.0 \cdot 10^{21} \text{ m}^{-3}$ (Fig. 1 left side). The "fine area" shows a quite homogeneous distribution of particles with sizes smaller than 5 nm. (Fig. 1b right side). The number density there was measured to $(1.5 \pm 1.0) \cdot 10^{23} \text{ m}^{-3}$ (up to ~ 100 times higher than in the "coarse area"). The thickness of the TEM film in the investigated area was measured using electron energy loss spectroscopy. The analysis of several TEM images show that about 2/3 of the material consists of the "fine" area and about 1/3 of the "coarse" area. The appearance of these two areas follows to a random pattern that does not reflect the grain structure of the material. It can be assumed that the areas with different distribution were formed as a result of Y_2O_3 distribution on powder particles during mechanical alloying.

The different nomenclatures were proposed by B. Odette [22] to distinguish between nanostructured ferritic alloys (NFAs) and oxide dispersion strengthened (ODS) steels. The main criterion for this distinguishing is the size distribution and number density of the

oxide particles. Accordingly, the 10–15 nm oxide particles as found for example in PM2000 steel [23] are called as “ODS particles” and finely dispersed oxides of a few nanometers in size and number density $>10^{23} \text{ m}^{-3}$ are called nano oxides (NO). Since both types of particles are present in the EUROFER-ODS alloy, the particles in the coarse region are referred to as “ODS particles” and the particles in the fine region are referred to as NO.

The existence of areas with different distribution of ODS particles, known as “bimodal distribution”, was already reported in a previous work [12,24,25]. Often, a bimodal distribution is associated with changes in grain size, where the “coarse area” is located in the grains smaller than 70 nm, while the formation of nano oxides in the “fine area” was observed in the large grains of 2–3 μm size [25]. Due to the significant influence of ODS particles distribution on the formation of radiation-induced defects, these two different areas will be referred to as “coarse area” and “fine area” in the following [12].

The large scale analysis of ODS particles showed that, depending on the position, 5% to 25% of the ODS particles in the “coarse” region have a (Ta,V,Y)O composition (Fig. 1c), while no such particles were observed in the fine region. This phase occurs as a separate particle or as a core in a complex particle typically surrounded by the Y_2O_3 phase. The (Ta,V,Y)O particles have a spherical shape very similar to that of Y_2O_3 particles, hence their identification on the basis of bright-field TEM images is practically impossible. The presence of Ta-rich oxide particles is the only visible difference between the coarse and the fine areas, suggesting that it may have played a role in the formation of the bimodal distribution. It can be assumed that based on the core-shell structure of some particles with (Ta,V,Y)O core and Y_2O_3 shell that the precipitation of the Ta-rich phase occurs on the earlier stages of hot isostatic pressing than that of Y_2O_3 particles. This initial formation of nucleation centers could promote the formation of ODS particles with sizes up to 30 nm. This is only an assumption, because so far there is no generally recognized explanation in the scientific literature for the bimodal size distribution of ODS particles in the same material class.

To the best of our knowledge, the formation of Ta-rich ODS particles has not been previously reported, despite an extensive characterization of the EUROFER-ODS alloy over 2 decades. The reasons for this are presumably the similar morphology for the one and nearly overlapping of Ta- $M\alpha,\beta$ (1.73–1.81keV) with W- $M\alpha,\beta$ (1.77–1.83keV) and Y- $L\alpha$ (1.92keV) EDX lines. The separation of Y and Ta was clearly possible by using the Ta- $L\alpha$ (8.1keV) and Y- $K\alpha$ (14.93keV) lines for the detection. The V-rich phase occurs in both areas. In the coarse area, it occurs as a shell around ODS particles and as nano-sized “eyes” on ODS particles (Fig. 1d). It is reasonable to assume that a thin V-rich layer has also formed around nano oxides in the “fine” area, however, clear imaging of such closely spaced, nano-sized, structures is below the sensitivity of analytical transmission electron microscopy.

3.2. Investigation of material irradiated at 300°C

The microstructure of the sample irradiated at 300°C is illustrated in Figs. 2–5. Fig. 2 marks the differences in defects formation inside “fine” and “coarse” areas. The WBDF image acquired using $g = [020]$ vector shows the defects in a grain oriented close to the $[102]$ zone-axis (Fig. 2a). The Y- and Cr-EDX maps visualize the distribution of ODS particles and M_{23}C_6 precipitates and allow an exact location of these both areas (Fig. 2b,c). The areas have the same crystallographic orientation and are thus parts of the same grain. This can be concluded from their identical g -conditions for all orientations. The “fine” area with one large Y_2O_3 particle (marked with the white arrow) is located in the center of the image, while the “coarse” area is located in its upper part (Fig. 2b). An average

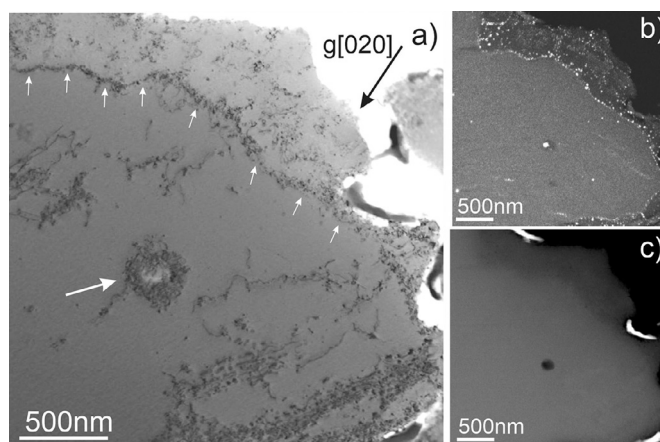


Fig. 2. Microstructure of EUROFER-ODS irradiated at 300°C. The Y map (b) and Cr (c) maps indicate the location of the “fine” and “coarse” areas as well as M_{23}C_6 precipitates.

line dislocation density of $(1.3 \pm 0.6) \cdot 10^{13} \text{ m}^{-2}$ and $(7 \pm 4) \cdot 10^{13} \text{ m}^{-2}$ was measured for the “fine” and “coarse” areas, respectively. However, there are variations in dislocation density by a factor of 2–5 for different grains.

The Burges vector of the radiation-induced defects and line dislocations was determined by the orientation of the grain near the $[111]$ and $[102]$ zone-axis as shown in Fig. 3 and Fig. 4. The g -vectors used for this purpose are listed in Table 1. Since the imaging with g -vectors of $\langle 110 \rangle$ type near $[111]$ zone axis does not allow a unambiguous determination of the Burgers vector for all defects, the sample was tilted to the zone-axis and imaged with $g[002]$ (Fig. 2a). It should be pointed out that all defects indicate the $b/2[11-1]$ direction of the Burger vector over wide areas, since 90% of them are invisible in the $g[1-10]$ images (Figs. 3c and Fig. 4c). Examination of several grains suggests that this preferential orientation of the Burger vector is frequently present in irradiated EUROFER-ODS steels.

The loops or point defects observed in the “fine” area tend to be adjacent to structural defects such as line dislocations, single Y_2O_3 particles (Fig. 2, 3), or grains between “coarse” and “fine” areas (Fig. 2a – marked with white arrows). A high density of radiation-induced defects, as visible for example in the lower part in Fig. 3, is the result of the higher local density of line dislocations. In contrast, the dislocation loops in the “coarse” area show a homogeneous spatial distribution (Fig. 4). The detailed quantitative analysis of size and number density of radiation induced defects for both areas are shown in the discussion section.

It was found that a single, separately located ODS particle (marked by arrow in Fig. 1a) can be found randomly in the “fine” area surrounded by nano oxides. The dislocation loops, as shown in Fig. 3, are also preferentially located close such to single ODS particle. They mainly have Burgers vector of the $\frac{1}{2}\langle 111 \rangle$ type. The analysis of each loop proved impossible due to its high local number density. A similar clustering of defects around ODS particles was also observed in neutron-irradiated EUROFER-ODS after irradiation at 250° [12].

Analyses of the acquired TEM images reveal that several line dislocations in the “fine” area have a tendency to transform into a helical dislocation. Fig. 5 presents several examples where the initial stage of its formation could be observed. The periodicity of the helix varied from 30 nm to 60 nm and its diameter from 5 nm to 15 nm. The maximal diameter of the helix observed in this study is about 30 nm (Fig. 4a).

The nano-sized voids or bubbles located on the ODS particles were observed in the “coarse” area. However, their appearance

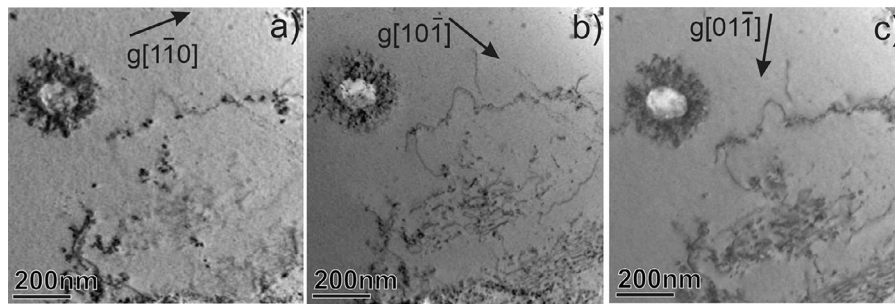


Fig. 3. The micrographs of the "fine area" obtained at WBDF conditions with selected g -vectors near $[111]$ ZA and imaged with reversal contrast.

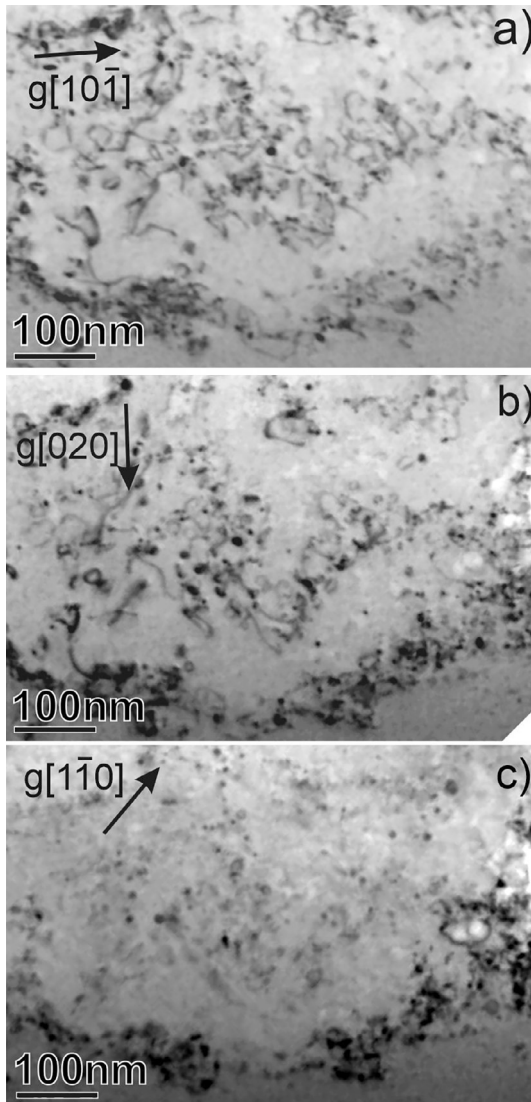


Fig. 4. The micrographs of the "coarse area" obtained at WBDF conditions with selected g -vectors near $[111]$ ZA and imaged with reversal contrast.

is irregular, so we excluded the voids from the detailed statistical analysis. The number density and size of the voids shown in Table 3 are a rough estimate based on analysis of several TEM micrographs.

3.3. Investigation of material irradiated at 400°C

Both dislocation loops and voids located on ODS particles were detected in the material irradiated at 400°C. The voids were

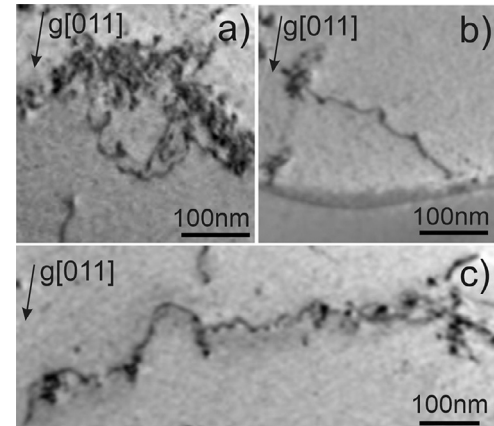


Fig. 5. Formation of helical dislocations in the "fine area".

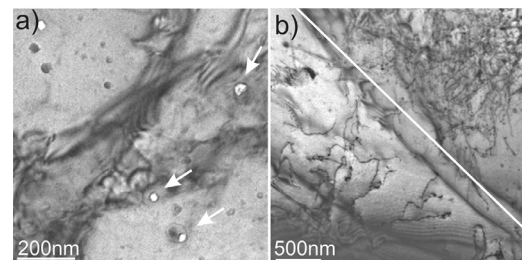


Fig. 6. The BF image of ODS particles with voids (a) and BF image illustrating the distribution of dislocations in the "coarse" (right side) and "fine" (left side) areas (b).

formed only in the "coarse" area and only at the ODS particles, as shown for example in Fig. 6a (marked by arrows). We have found no evidence for their existence in the "fine" area, but we cannot completely exclude it, since nanoscale (~ 1 nm) voids cannot be clearly distinguished from nano oxides. Their occurrence is not homogeneous i.e., in some "coarse" areas voids were found on each ODS particle, in some other areas with ODS particles no voids were found.

The BF image in Fig. 6b illustrates typical microstructural differences between "fine" (bottom left side) and "coarse" (upper right side) areas, with the white line indicating approximately the boundary between them. The density of line dislocations was determined as $(4 \pm 1.5) \cdot 10^{13} \text{ m}^{-2}$ in the "fine" and as $(1.6 \pm 0.7) \cdot 10^{14} \text{ m}^{-2}$ in the "coarse" area (see Table 3). Analyses of several grains show that the "coarse" areas have a systematically 3–5 times increased dislocation density compared to the "fine" areas. In the "coarse" area, the formation of $b(100)$ dislocation junctions formed by the interaction of two $b\frac{1}{2}(111)$ non-coplanar line dislocations have often been observed. Since the junctions with $b(100)$ were typically detected in the plastically deformed material [26],

Table 3

Quantitative and statistical data of dislocation loops and voids obtained separately for "fine" and "coarse" areas.

T_{irr}	area	density of line dislocations / $\times 10^{13} \text{ m}^{-2}$	loop number density / $\times 10^{21} \text{ m}^{-3}$	average size /nm	void number density / $\times 10^{19} \text{ m}^{-3}$	void swelling %	relaxation volume %	fraction of $\frac{1}{2}\langle 111 \rangle$ loops
300°C	fine	1.3 ± 0.6	1.0 ± 0.7	7	---	---	0.008 ± 0.005	100%
	coarse	7 ± 4	5.0 ± 1.5	13	0.5 ± 0.4	~ 0.004	0.035 ± 0.015	95%
400°C	fine	4.0 ± 1.5	0.45 ± 0.20	71	---	---	0.035 ± 0.02	90%
	coarse	17 ± 8	1.0 ± 0.5	94	3.0 ± 2.0	~ 0.02	0.08 ± 0.03	85%

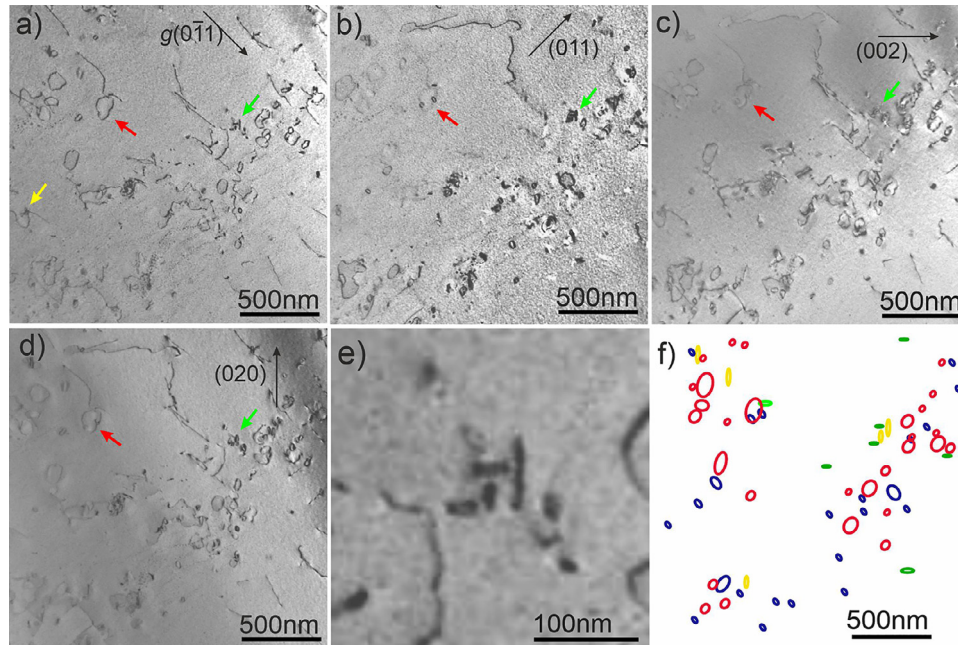


Fig. 7. The micrographs obtained at WBDF conditions with selected g -vectors near $[100]$ ZA (parts a-d). The part (e) shows the presence of $b(100)$ loops. In part (f), loops with $b[1-11]$ and $b[111]$ Burgers vectors are schematically drawn with red and blue colors, and $b[010]$ and $b[001]$ loops are drawn with green and yellow colors, respectively.

their occurrence in the "coarse" area could be considered as an indication of a radiation-induced movement of the line dislocations. The fact that in unirradiated EUROFER-ODS no junctions were observed additionally confirms this assumption.

Due to the high dislocation density in the coarse areas, we omitted a detailed statistical evaluation of the dislocation loops there. It was observed that the larger loops were often cut off by the foil surface. Such loops are often visible as typical line dislocations [3]. This effect also leads to a significant distortion of the loop statistics.

The WBDF imaging with different g -vectors near the $[100]$ zone-axis was applied to determine the Burgers vector of dislocation loops (Fig. 7). In the parts (a)-(d) are shown two beam images with all four low indexed g -vectors. The Burgers vector of the loops were identified based on their projections and visible criteria. The 90% of loops have a $b\frac{1}{2}\langle 111 \rangle$ Burgers vector, while only a few $b(100)$ loops were observed. The accumulation of $b(100)$ loops is indicated by a green arrow in g -images and displayed enlarged in Fig. 7e. The $b(100)$ loops are frequently found as individual loops, so that such "agglomerations" of 5 loops in a limited space are rather exceptional. The maximum size of the $b(100)$ loops is about 30 nm, which is considerably smaller than for the $b\frac{1}{2}\langle 111 \rangle$ loops, whose size can exceed 120 nm. Fig. 7f summarizes the results of Burgers vector analysis, where the loops are imaged using different colors according to their respective Burgers vector.

Despite the low number density of $4.5 \times 10^{20} \text{ m}^{-3}$, the loops often interact with each other and with the line dislocations. Exam-

ples of such interactions are shown in Figs. 8 and 9. Fig. 8 presents a detailed analysis of interacting $b\frac{1}{2}\langle 111 \rangle$ loops with different Burgers vector directions. The 150 nm sized loop with $b\frac{1}{2}[1-11]$ (red color) interacts with two 20 nm sized loops having a $b\frac{1}{2}[111]$ Burgers vector (blue color). The colors used for the labeling the Burgers vector were selected according to Table 2. The shape of the large loops shows a deformation at the points where it touches the smaller loops (Fig. 8a). The invisibility of the contact position in Fig. 9d indicates the formation of a junction having a $b[001]$ Burgers vector.

Fig. 9 shows the interaction of the line dislocation with a loop, where the line dislocation has a characteristic bow. Such structures are frequently observed in the sample. Both the bow and the line dislocation have the same Burgers vector of type $b\frac{1}{2}\langle 111 \rangle$ and the same direction. It can be assumed that the line dislocation moves in the habitat plane of the dislocation loops and causes an annihilation of the loops. This process explains the low number density of dislocation loops of $4.5 \times 10^{20} \text{ m}^{-3}$ being present in this area.

4. Discussion

The microstructural response of different ODS-based alloys to neutron or ion irradiation has been extensively studied in the past using TEM or atom probe tomography (APT) methods [12,27,28]. Typically, the microstructural examination of ODS materials focused on the radiation effect on the size, distribution, and chemical composition of ODS particles [11]. The formation of radiation-induced defects such as loops or voids in ODS materials has rarely

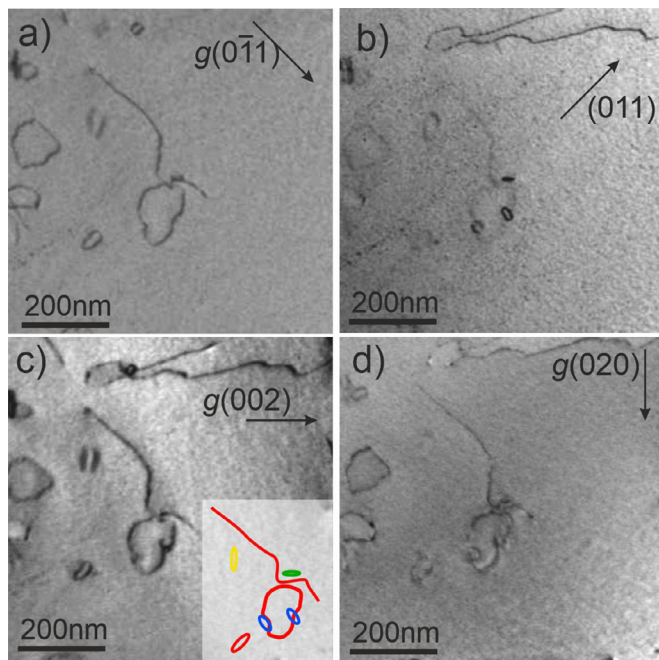


Fig. 8. The interaction of several $b\frac{1}{2}\langle 111 \rangle$ loops imaged with different g -vectors near $[100]$ zone axis. The loops are drawn in part (c) with colors according to the Table 2: $b\frac{1}{2}[1-11]$ red, $b\frac{1}{2}[111]$ blue, $b[002]$ yellow and $b[020]$ green.

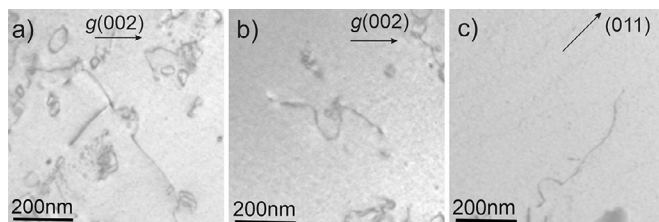


Fig. 9. The micrographs show a possible interaction of line dislocations with dislocation loops.

been the subject of research. The formation of small point defects, "black dots", was analyzed for example in neutron irradiated EUROFER-ODS alloys up to a dose of 3 dpa at 300°C, 450°C and 550°C [8]. At present time, 300°C is the lowest irradiation temperature for ODS steels being reported in scientific literature. The stability of ODS particles under neutron irradiation and the low formation threshold for dislocation structures have been reported for irradiation temperatures above 400°C [8,29]. This limited amount of published data and the unequal irradiation conditions make a direct comparison of our own results with the scientific literature at least difficult or even impossible.

Table 3 summarizes all relevant statistical and quantitative data on dislocation loops and voids extracted by analyzing, among others, the demonstrated TEM images. The data includes average size, number density, Burgers vector, fraction of $b\frac{1}{2}\langle 111 \rangle$ loops and relaxation volume. The total number density of radiation-induced defects was obtained by taking into account that part of the defects is invisible at defined g -conditions (Table 1,2). In the images with $g(200)$, 100% of $b\frac{1}{2}\langle 111 \rangle$ loops and only 1/3 of $b\langle 100 \rangle$ loops are visible. Correspondingly, 50% of $b\frac{1}{2}\langle 111 \rangle$ loops and 66% of $b\langle 001 \rangle$ loops are visible in the images with $g(011)$. The average of different g -images from several grains was determined as the total number density value of dislocation loops.

As shown in several publications, the ODS particles serve as a sink for vacancies or dissolved gas atoms, leading to the formation of bubbles/cavities directly on the particle surfaces, and thus pre-

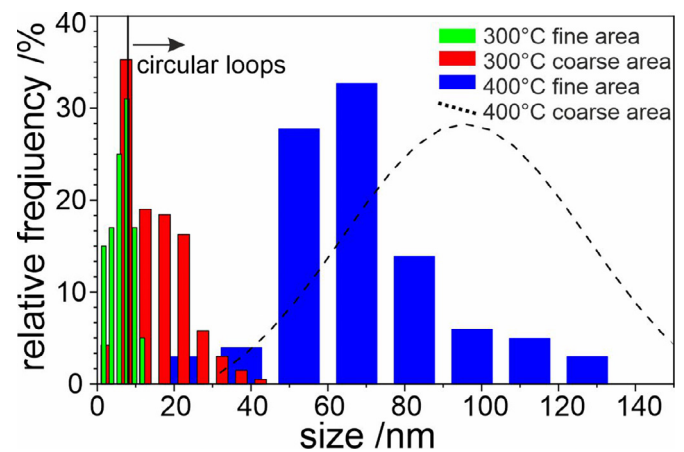


Fig. 10. Size distribution histograms of dislocation loops for 300°C and 400°C irradiation temperatures.

venting their influence on the mechanical properties [10,12,30,31]. Experimentally, this fact was confirmed in both ion and neutron irradiated ODS alloys. In the present study, the voids located on the ODS particles were formed only in the coarse area at both irradiation temperatures (Fig. 6). However, their appearance is irregular, so that a number density of $\sim 10^{19} \text{m}^{-3}$ was estimated with considerable error of 70%-90% (Table 3). The reasons for this are the irregular void formation and the estimation of the occurrence of the "fine" and "coarse" areas. The existence of nanometer-sized voids in the "fine" area cannot be definitely excluded, since they would be indistinguishable from nano oxides in TEM images. Since no voids were found in the fine area after irradiation at 250°C, 350°C and 450°C, it can be assumed that the presence of nano oxides effectively prevents (or at least minimizes) their formation in a broad temperature range [12]. Radiation-induced formation of voids in ferritic-martensitic steels, including ODS alloys has typically been observed after neutron irradiation in a narrow temperature range from 300°C to 450°C [2,3,32]. It was also shown that the effect of neutron irradiation on microstructure is strongest in this particular temperature range. The presence of He or H in the matrix plays an important role in the void formation process i.e., it increases the void swelling by 3-5 times [31,32].

The size distribution histograms of the dislocation loops in both specimens are displayed in different colors in Fig 10. The radiation-induced defects formed at 300°C in the "fine area" show a narrow size distribution with sizes of less than 10 nm (green bars). The majority of them are referred to as "black dots", which have sizes less than ~ 7 nm. In the TEM images such loops show a dot-like projection without a recognizable structure, whereas the larger loops usually show a circular structure in the images. The separation line located at 7 nm in the diagram shows approximately the difference between "black dots" and clearly visible circular loops. In general, the size at which the loops are imaged as rings depends strongly on the sample thickness, alignment and indexing of the g -vector and other imaging conditions. The loops in the coarse area show a broad distribution with sizes up to 30 nm and a significantly higher number density.

The loops formed at 400°C in the "fine" area show a broad size distribution ranging from 20 nm to 130 nm with an average value of 65 nm. The distribution in the "coarse" area has been roughly estimated and drawn using a dashed line instead of a bar histogram. A significant fraction of the loops has a size larger than 100 nm, exceeding the typical thickness of a TEM foil. Such large loops are often cut off by the foil surface and displayed as bows. Their identification in TEM images was only possible if at least 1/3 of the original circle is visible. The high density of line dislocations

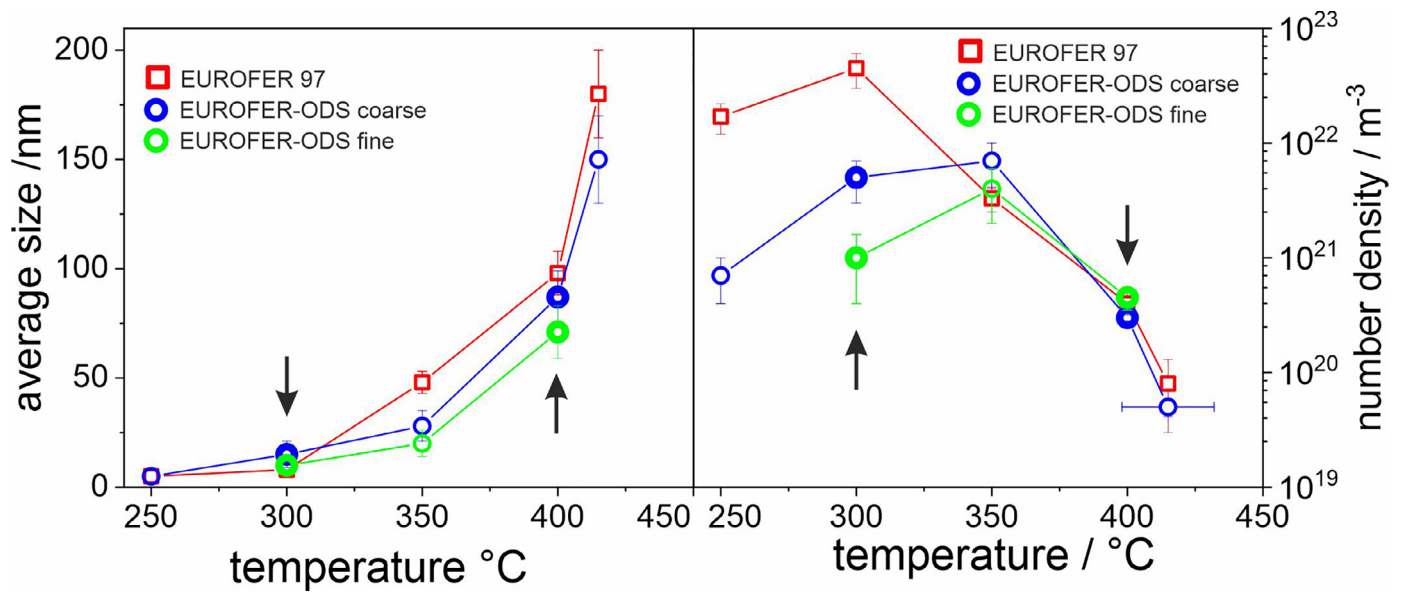


Fig. 11. Dependence of average size (a) and number density (b) of dislocation loops in EUROFER-97 [3] (red squares) and EUROFER-ODS [12] (blue and green circles) on irradiation temperature.

in the coarse area is an additional factor affecting the statistics. It can also be assumed that a considerable number of the larger loops were imaged as line dislocations and not detected as loop or "radiation-induced defects".

Quantitative data on dislocation loops were compared with EUROFER97 and EUROFER-ODS from the entire SPICE irradiation program [3,12] (Fig. 11). The data described in this publication are drawn with a thick symbol and additionally marked with black arrows. The average size and number density from the "fine" range in EUROFER-ODS were both plotted only for 300°C, 350°C, and 400°C (green circles), since the no loops were observed at 250°C and 415°C [12]. The average size of the loops is almost identical for both alloys at 250°C and 300°C, while at higher temperatures it appears systematically ~20–35% smaller in ODS material than in EUROFER97 (Fig. 11a).

The number density of dislocation loops shows a significant difference between both alloys at an irradiation temperature of 250°C and 300°C (Fig. 11b), respectively. Their value is one order of magnitude higher in EUROFER97 than in the "coarse" area of the EUROFER-ODS. However, at higher irradiation temperatures, the number densities of both EUROFER and EUROFER-ODS are comparable. The highest number density was measured for EUROFER-ODS irradiated at 350°C, while for EUROFER97 it reaches its maximum at 300°C. In summary it can be noted that the loops in the "fine" area are systematically smaller in size and have a lower number density than in the "coarse" area. It is assumed that the suppression of radiation damage by the ODS particles is only effective up to 350°C. At higher temperatures, the formation of dislocation loops is fairly independent of their presence and morphology.

The growth of interstitial dislocation loops during irradiation results from the interstitial elastic misfit, which forces dislocations to attract interstitials stronger than vacancies [33]. A dislocation loop is then formed by inserting a circular plane of atoms between closely packed atomic planes. The additional volume introduced this plane is referred to as relaxation volume the dislocation loop [34]. Accordingly, the relaxation volume Ω can be calculated as the volume of the prism by $\Omega = b \cdot A$ expression, where b is the Burgers vector and A is the surface area of the dislocation loop. The values of the Burgers vectors for $b_{\frac{1}{2}\langle 111 \rangle}$ and $b_{\langle 100 \rangle}$ loops were calculated as $b_{\frac{1}{2}\langle 111 \rangle} = \frac{1}{2}a_0\sqrt{3} = 0.247 \text{ nm}$ and $b_{\langle 100 \rangle} = a_0 = 0.286 \text{ nm}$, respectively.

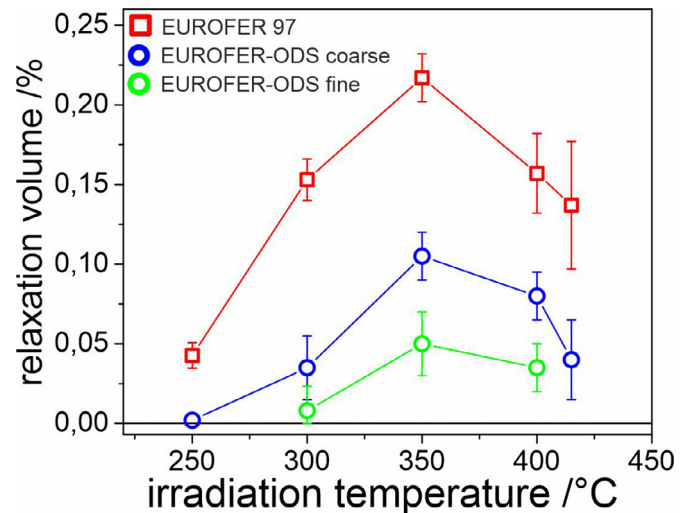


Fig. 12. Relaxation volume of interstitial dislocation loops formed in EUROFER97 and EUROFER-ODS as a function of radiation temperature.

The diagram plotted in Fig. 12 shows the dependence of the relaxation volume of the interstitial loops in EUROFER97 and EUROFER-ODS on the irradiation temperatures. It is noticeable that the cumulative relaxation volume in EUROFER-ODS is systematically smaller than in EUROFER97, which was irradiated under the same conditions. The maximum value was measured at 350°C for both alloys (Fig. 12, Table 3). At temperatures exceeding 400°C, the measurement error increases significantly due to the difficulty in identifying the loops and the uncertainty in size determination. Those loops larger than the thickness of the TEM film are often visible as line dislocations [3]. It should be noticed that information on dislocation loops formed at higher temperatures is scarce. Yao et al. [15] reported loops with sizes exceeding 150 nm in pure single crystal iron after ion irradiation at 500°C. Theoretical studies predict that continuous irradiation causes the structural transition from an ensemble of isolated defects to a microstructure dominated by a dislocation network [35]. The formation of a dense dislocation network in the coarse area was also observed within the present work and in [3]. It was also clearly shown that the forma-

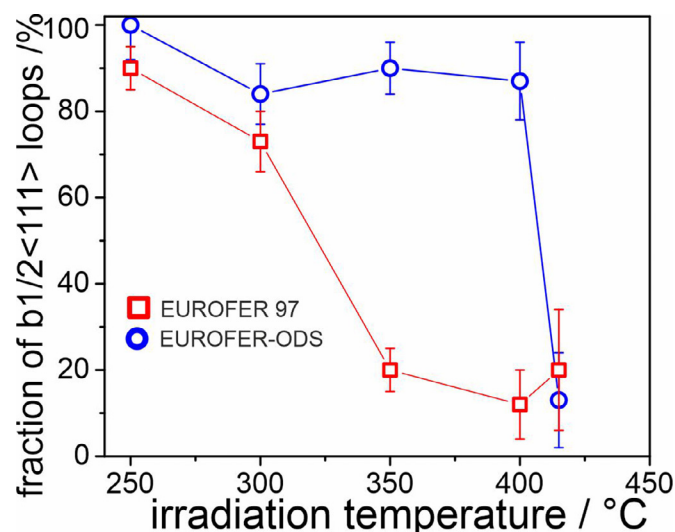


Fig. 13. Estimated fraction of dislocation loops of the $b\frac{1}{2}\langle 111 \rangle$ type for EUROFER97 (red squares) and EUROFER-ODS (blue circles).

tion of loops also occurred at 420°C in EUROFER97 and EUROFER-ODS but they are not always recognized as loops by TEM (Fig. 6b).

The dependence of the Burgers vector of the loops in EUROFER97 and EUROFER-ODS on the irradiation temperature is shown in a diagram in Fig. 13. At 250°C and 300°C irradiation, the $\frac{1}{2}\langle 111 \rangle$ -loops dominate in both alloys. At higher temperatures, the $\langle 100 \rangle$ loops form preferentially in EUROFER97, whose percentage increases to over 80%. The $\frac{1}{2}\langle 111 \rangle$ loops remain in EUROFER-ODS as the dominant defect type for all higher irradiation temperatures except the highest, i.e., 420°C. The preferential formation of $\frac{1}{2}\langle 111 \rangle$ loops was also observed in ion-irradiated EUROFER-ODS ($T_{irr}=400^\circ\text{C}$), while in EUROFER97 the majority of loops were of the $\langle 111 \rangle$ type [36].

The current results imply that ODS alloys with a higher number density of "fine" nano oxides have a higher radiation resistance than alloys with "coarse" ODS particles and a low number density. The differences between "coarse" and "fine" areas are evident at all irradiation temperatures. At 350°C, the damage reaches a maximum value i.e., increased formation of voids and dislocation loops (Fig. 11b, Fig. 12). This effect was also observed in boron-alloyed EUROFER97 [32] and other materials irradiated under the similar irradiation conditions [2].

Numerous ODS alloys were developed and fabricated during past decade. The number density of nano-sized particles varied by about two orders of magnitude from $\sim 10^{23} \text{ m}^{-3}$ in nanostructured ferritic alloys [22] down to $\sim 10^{21} \text{ m}^{-3}$ in ODS alloys of PM2000 type [30] or EUROFER-ODS [12,27]. The presence of highly dispersed oxide particles effectively suppresses the formation of radiation-induced defects and thus minimizes their effect on mechanical properties such as hardening. This has been observed in ferritic-martensitic 9-10Cr ODS steels (such as EUROFER-ODS) and nanoscale ferritic 13-17Cr ODS steels (such as 14YWT) [37,38]. Although this important observation has not yet been analyzed quantitatively in existing literature, e.g., by computational materials science. It now seems qualitatively obvious that radiation-induced loops are unlikely to form while the existing sink force density is exceeding a certain threshold value. This means that, when the number density of nanoscale oxide particles (e.g. Y_2O_3 , $\text{Y}_2\text{Ti}_2\text{O}_5$) significantly exceeds the loop number density in conventional particle-free cast steel irradiated under the same conditions. The critical number density of ODS particles is $\sim 10^{22} \text{ m}^{-3}$ for an irradiation temperature of 250-300°C. In alloys with higher number density of ODS particles, irradiation-induced Frenkel de-

fects do not contribute with high probability to nucleation and growth of stable loops, but rather are annihilated by trapping at the ODS-particle matrix interface. Homogeneously dispersed, nano oxides have therefore an outstanding additional function: They are not only the foundation for high-temperature creep and fatigue strength but also remarkably reduce irradiation hardening and consequently retard substantially irradiation embrittlement at lower strain rates relevant for tensile and creep experiments.

5. Conclusions

The study presents a detailed microstructural examination of neutron irradiated EUROFER-ODS at 300°C and 400°C. This involves imaging of radiation-induced defects under weak beam conditions and their quantitative analysis, which includes determination of their Burger vector, nature, morphology and spatial distribution. The presence of ODS particles inside EUROFER-ODS generally minimize the formation of radiation-induced defects, as shown by comparison with the otherwise identical alloy EUROFER97. The finely dispersed nano oxides with a number density higher than 10^{23} m^{-3} show a significantly stronger effect than larger ODS particles with lower number density. This is particularly the case for irradiation temperatures of 300°C and below. Despite the fact that the highest number density of radiation-induced defects was measured at $<300^\circ\text{C}$, the largest effect on the microstructure, measured by the cumulative volume of dislocation loops, was determined to be at 350°C. The microstructure was found to be dominated by interstitial loops with mainly $\frac{1}{2}\langle 111 \rangle$ Burgers vector.

The results were supplemented with data from EUROFER-ODS irradiated at other temperatures [12] and compared with EUROFER97 irradiated as the part of the same irradiation campaign [3]. This analysis provides the opportunity to examine the influence of ODS particles on the formation of radiation-induced defects in the temperature range from 250°C to 450°C.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work has been carried out within the framework of the EURO fusion Consortium and has received funding from the Euratom research and training programme 2014-2018 and 2019-2020 under grant agreement No. 633053. The views and opinions expressed herein do not necessarily reflect those of the European Commission.

References

- [1] J. Knaster, A. Moeslang, T. Muroga, Materials research for fusion, *Nature Phys.* 12 (2016) 424–434, doi:10.1038/nphys3735.
- [2] A. Kimura, Current status of reduced-activation ferritic/martensitic steels R&D for Fusion Energy, *Mater. Trans.* 46 (2005) 394–404, doi:10.2320/MATERTRANS.46.394.
- [3] M. Klimenkov, U. Jäntschi, M. Rieth, A. Möslang, Correlation of microstructural and mechanical properties of neutron irradiated EUROFER97 steel, *J. Nucl. Mater.* 538 (2020) 152231, doi:10.1016/j.jnucmat.2020.152231.
- [4] E. Aydogan, J.S. Weaver, U. Carvajal-Nunez, M.M. Schneider, J.G. Gigax, D.L. Krumwiede, P. Hosemann, T.A. Saleh, N.A. Mara, D.T. Hoelzer, B. Hilton, S.A. Maloy, Response of 14YWT alloys under neutron irradiation: a complementary study on microstructure and mechanical properties, *Acta Mater.* 167 (2019) 181–196, doi:10.1016/j.actamat.2019.01.041.
- [5] J. Wang, M.B. Toloczko, V.N. Voevodin, V.V. Bryk, O.V. Borodin, V.V. Mel'nychenko, A.S. Kalchenko, F.A. Garner, L. Shao, Atom probe tomography characterization of high-dose ion irradiated MA957, *J. Nucl. Mater.* 545 (2021) 152528, doi:10.1016/j.jnucmat.2020.152528.

- [6] T. Gräning, M. Klimenkov, M. Rieth, C. Heintze, A. Möslang, Long-term stability of the microstructure of austenitic ODS steel rods produced with a carbon-containing process control agent, *J. Nucl. Mater.* 523 (2019) 111–120, doi:[10.1016/j.jnucmat.2019.05.060](https://doi.org/10.1016/j.jnucmat.2019.05.060).
- [7] S. Chen, Y. Wang, K. Tadaki, N. Hashimoto, S. Ohnuki, Suppression effect of nano-sized oxide particles on helium irradiation hardening in F82H-ODS steel, *J. Nucl. Mater.* 455 (2014) 301–305, doi:[10.1016/j.jnucmat.2014.06.041](https://doi.org/10.1016/j.jnucmat.2014.06.041).
- [8] M. Kolluri, P.D. Edmondson, N.V. Luzginova, F. Berg, A structure-property correlation study of neutron irradiation induced damage in EU batch of ODS Eurofer97 steel, *Mater. Sci. Eng.: A* 597 (2014) 111–116, doi:[10.1016/j.msea.2013.12.074](https://doi.org/10.1016/j.msea.2013.12.074).
- [9] C. Heintze, F. Bergner, M. Hernández-Mayoral, R. Kögler, G. Müller, A. Ulbricht, Irradiation hardening of Fe-9Cr-based alloys and ODS Eurofer: Effect of helium implantation and iron-ion irradiation at 300°C including sequence effects, *J. Nucl. Mater.* 470 (2016) 258–267, doi:[10.1016/j.jnucmat.2015.12.041](https://doi.org/10.1016/j.jnucmat.2015.12.041).
- [10] A.I. Ryazanov, O.K. Chugunov, S.M. Ivanov, S.T. Latushkin, R. Lindau, A. Möslang, A.A. Nikitina, K.E. Prikhodko, E.V. Semenov, V.N. Unezhev, P.V. Vladimirov, Tensile properties and microstructure of helium implanted EUROFER ODS, *J. Nucl. Mater.* 442 (2013) S153–S157, doi:[10.1016/j.jnucmat.2013.03.080](https://doi.org/10.1016/j.jnucmat.2013.03.080).
- [11] J.P. Wharry, M.J. Swenson, K.H. Yano, A review of the irradiation evolution of dispersed oxide nanoparticles in the b.c.c. Fe-Cr system: Current understanding and future directions, *J. Nucl. Mater.* 486 (2017) 11–20, doi:[10.1016/j.jnucmat.2017.01.009](https://doi.org/10.1016/j.jnucmat.2017.01.009).
- [12] M. Klimenkov, R. Lindau, U. Jäntschi, A. Möslang, Effect of irradiation temperature on microstructure of ferritic-martensitic ODS steel, *J. Nucl. Mater.* 493 (2017) 426–435, doi:[10.1016/j.jnucmat.2017.06.024](https://doi.org/10.1016/j.jnucmat.2017.06.024).
- [13] E. Materna-Morris, R. Lindau, H.-C. Schneider, A. Möslang, Tensile behavior of EUROFER ODS steel after neutron irradiation up to 16.3 dpa between 250 and 450°C, *Fusion Eng. Desig.* 98–99 (2015) 2038–2041, doi:[10.1016/j.fusengdes.2015.07.015](https://doi.org/10.1016/j.fusengdes.2015.07.015).
- [14] N. Akasaka, S. Yamashita, T. Yoshitake, S. Ukai, A. Kimura, Microstructural changes of neutron irradiated ODS ferritic and martensitic steels, *J. Nucl. Mater.* 329–333 (2004) 1053–1056, doi:[10.1016/j.jnucmat.2004.04.133](https://doi.org/10.1016/j.jnucmat.2004.04.133).
- [15] Z. Yao, M.L. Jenkins, M. Hernández-Mayoral, M.A. Kirk, The temperature dependence of heavy-ion damage in iron: a microstructural transition at elevated temperatures, *Philos. Mag.* 90 (2010) 4623–4634, doi:[10.1080/14786430903430981](https://doi.org/10.1080/14786430903430981).
- [16] H.J. Jung, D.J. Edwards, R.J. Kurtz, T. Yamamoto, Y. Wu, G.R. Odette, Structural and chemical evolution in neutron irradiated and helium-injected ferritic ODS PM2000 alloy, *J. Nucl. Mater.* 484 (2017) 68–80, doi:[10.1016/j.jnucmat.2016.11.022](https://doi.org/10.1016/j.jnucmat.2016.11.022).
- [17] N. Baluc, R. Schäublin, P. Spätig, M. Victoria, On the potentiality of using ferritic/martensitic steels as structural materials for fusion reactors, *Nucl. Fusion* 44 (2004) 56–61, doi:[10.1088/0029-5515/44/1/006](https://doi.org/10.1088/0029-5515/44/1/006).
- [18] E. Materna-Morris, A. Möslang, H.-C. Schneider, Tensile and low cycle fatigue properties of EUROFER97-steel after 16.3 dpa neutron irradiation at 523, 623 and 723K, *J. Nucl. Mater.* 442 (2013) S62–S66, doi:[10.1016/j.jnucmat.2013.03.038](https://doi.org/10.1016/j.jnucmat.2013.03.038).
- [19] M. Klimiankou, R. Lindau, A. Möslang, Direct correlation between morphology of (Fe,Cr)23C6 precipitates and impact behavior of ODS steels, *J. Nucl. Mater.* 367–370 (2007) 173–178, doi:[10.1016/j.jnucmat.2007.03.150](https://doi.org/10.1016/j.jnucmat.2007.03.150).
- [20] R. Schaublin, T. Leguey, P. Spätig, N. Baluc, M. Victoria, Microstructure and mechanical properties of two ODS ferritic/martensitic steels, *J. Nucl. Mater.* 307–311 (2002) 778–782, doi:[10.1016/S0022-3115\(02\)01193-5](https://doi.org/10.1016/S0022-3115(02)01193-5).
- [21] R. Lindau, M. Klimenkov, U. Jäntschi, A. Möslang, L. Commin, Mechanical and microstructural characterization of electron beam welded reduced activation oxide dispersion strengthened – Eurofer steel, *J. Nucl. Mater.* 416 (2011) 22–29, doi:[10.1016/j.jnucmat.2011.01.025](https://doi.org/10.1016/j.jnucmat.2011.01.025).
- [22] G.R. Odette, Recent progress in developing and qualifying nanostructured ferritic alloys for advanced fission and fusion applications, *JOM* 66 (2014) 2427–2441, doi:[10.1007/s11837-014-1207-5](https://doi.org/10.1007/s11837-014-1207-5).
- [23] M. Klimenkov, A. Möslang, R. Lindau, EELS analysis of complex precipitates in PM 2000 steel, *Eur. Phys. J. Appl. Phys.* 42 (2008) 293–303, doi:[10.1051/EPJAP:2008084](https://doi.org/10.1051/EPJAP:2008084).
- [24] C. Eisel, M. Klimenkov, R. Lindau, A. Möslang, H. Sandim, A.F. Padilha, D. Raabe, High-resolution transmission electron microscopy and electron backscatter diffraction in nanoscaled ferritic and ferritic-martensitic oxide dispersion strengthened-steels, *J. Nucl. Mater.* 385 (2009) 231–235, doi:[10.1016/j.jnucmat.2008.11.029](https://doi.org/10.1016/j.jnucmat.2008.11.029).
- [25] P. He, M. Klimenkov, R. Lindau, A. Möslang, Characterization of precipitates in nano structured 14% Cr ODS alloys for fusion application, *J. Nucl. Mater.* 428 (2012) 131–138, doi:[10.1016/j.jnucmat.2011.08.026](https://doi.org/10.1016/j.jnucmat.2011.08.026).
- [26] M. Stricker, D. Weygand, Dislocation multiplication mechanisms – Glissile junctions and their role on the plastic deformation at the microscale, *Acta Mater.* 99 (2015) 130–139, doi:[10.1016/j.actamat.2015.07.073](https://doi.org/10.1016/j.actamat.2015.07.073).
- [27] R. Schäublin, A. Ramar, N. Baluc, V. de Castro, M.A. Monge, T. Leguey, N. Schmid, C. Bonjour, Microstructural development under irradiation in European ODS ferritic/martensitic steels, *J. Nucl. Mater.* 351 (2006) 247–260, doi:[10.1016/j.jnucmat.2006.02.005](https://doi.org/10.1016/j.jnucmat.2006.02.005).
- [28] S. Rogozhkin, A. Bogachev, O. Korchuganova, A. Nikitin, N. Orlov, A. Aleev, A. Zaluzhnyi, M. Kozodaev, T. Kulevoy, B. Chalykh, R. Lindau, J. Hoffmann, A. Möslang, P. Vladimirov, M. Klimenkov, M. Heilmair, J. Wagner, S. Seils, Nanostructure evolution in ODS steels under ion irradiation, *Nucl. Mater. Energy* 9 (2016) 66–74, doi:[10.1016/j.nme.2016.06.011](https://doi.org/10.1016/j.nme.2016.06.011).
- [29] M.J. Swenson, J.P. Wharry, The comparison of microstructure and nanocluster evolution in proton and neutron irradiated Fe-9%Cr ODS steel to 3 dpa at 500°C, *J. Nucl. Mater.* 467 (2015) 97–112, doi:[10.1016/j.jnucmat.2015.09.022](https://doi.org/10.1016/j.jnucmat.2015.09.022).
- [30] M. Klimiankou, R. Lindau, A. Möslang, Energy-filtered TEM imaging and EELS Study of ODS particles and argon-filled cavities in ferritic-martensitic steels, *Micron* (Oxford, England 1993) 36 (2005) 1–8, doi:[10.1016/j.micron.2004.08.001](https://doi.org/10.1016/j.micron.2004.08.001).
- [31] N. Zimmer, P. Vladimirov, M. Klimenkov, U. Jäntschi, R. Vila, V. Chakin, F. Mota, Microstructural evolution of three potential fusion candidate steels under ion-irradiation, *J. Nucl. Mater.* 535 (2020) 152160, doi:[10.1016/j.jnucmat.2020.152160](https://doi.org/10.1016/j.jnucmat.2020.152160).
- [32] M. Klimenkov, A. Möslang, E. Materna-Morris, Helium influence on the microstructure and swelling of 9%Cr ferritic steel after neutron irradiation to 16.3 dpa, *J. Nucl. Mater.* 453 (2014) 54–59, doi:[10.1016/j.jnucmat.2014.05.001](https://doi.org/10.1016/j.jnucmat.2014.05.001).
- [33] N.A. Smallman R.E., *Modern Physical Metallurgy*, Elsevier, 2014.
- [34] S.L. Dudarev, P.-W. Ma, Elastic fields, dipole tensors, and interaction between self-interstitial atom defects in bcc transition metals, *Phys. Rev. Mater.* 2 (2018), doi:[10.1103/PhysRevMaterials.2.033602](https://doi.org/10.1103/PhysRevMaterials.2.033602).
- [35] P.M. Derlet, S.L. Dudarev, Microscopic structure of a heavily irradiated material, *Phys. Rev. Mater.* 4 (2020), doi:[10.1103/PhysRevMaterials.4.023605](https://doi.org/10.1103/PhysRevMaterials.4.023605).
- [36] D. Brimbal, L. Beck, O. Troeber, E. Gaganidze, P. Trocellier, J. Aktaa, R. Lindau, Microstructural characterization of Eurofer-97 and Eurofer-ODS steels before and after multi-beam ion irradiations at JANNUS Saclay facility, *J. Nucl. Mater.* 465 (2015) 236–244, doi:[10.1016/j.jnucmat.2015.05.045](https://doi.org/10.1016/j.jnucmat.2015.05.045).
- [37] D.A. McClintock, M.A. Sokolov, D.T. Hoelzer, R.K. Nanstad, Mechanical properties of irradiated ODS-EUROFER and nanocluster strengthened 14YWT, *J. Nucl. Mater.* 392 (2009) 353–359, doi:[10.1016/j.jnucmat.2009.03.024](https://doi.org/10.1016/j.jnucmat.2009.03.024).
- [38] S.J. Zinkle, A. Möslang, T. Muroga, H. Tanigawa, Multimodal options for materials research to advance the basis for fusion energy in the ITER era, *Nucl. Fusion* 53 (2013) 104024, doi:[10.1088/0029-5515/53/10/104024](https://doi.org/10.1088/0029-5515/53/10/104024).